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Copernicus Atmosphere Monitoring Service

CAMS_81 – Global and Regional emissions

D81.2.2.4 : CAMS-GLOB-ANT emissions for all species for year 2019, and CAMS-GLOB-AIR emissions for 2000-2019

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Contributors

CNRS-LA
C. Granier, N. Elguindi

CNRS-OMP
S. Darras

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1. Summary

This document describes the CAMS global anthropogenic emissions (CAMS-GLOB-ANT_v2.1) for 2019 and the CAMS aircraft emissions (CAMS-GLOB-AIR) for 2000-2019, i.e. the methodology, documentation and first results. The dataset will be released at the end of February 2019 as planned in the contract.

As described in Deliverables D81.5.1.1-M6, D81.5.2.1-M6, D81.5.3.1-M6, D81.5.4.1, the global emissions dataset will be delivered through the ECCAD database, in a restricted area of ECCAD: they will be available by mid-March 2019.

Once validated and published, the CAMS-81 emissions will become fully public.

2. Methodology for developing the 2019 global anthropogenic emissions

The strategy to develop the CAMS global emissions as proposed in the contract is shown in Figure 1, for the 1st year of the project. It was already detailed in D81.2.2.

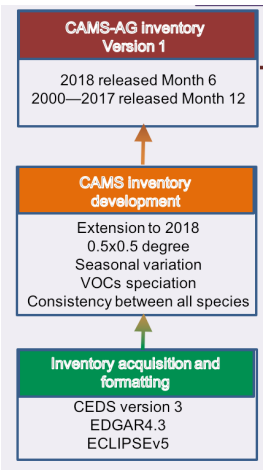


Figure 1: Methodology for developing the anthropogenic global emissions

The CAMS-81 2019 emissions are based on two datasets, EDGAR4.3 developed by the European Joint Center (Crippa et al., ACP, 2016) and the CEDS emissions (Smith et al., 2015) which provide emissions for the next IPCC report, AR6. Characteristics of these datasets are shown in Table 1.

Name of inventory	Species considered	Period covered	Spatial resolution
EDGAR4.3	BC, OC, NOx, NH3, SO2, NMVOCs, CO	1970-2010	0.1x0.1 degree
	CO2 (from EDGAR4.3.2)	1970-2012	
	CH4 and N2O (from EDGAR4.3.2)	1970-2012	
CEDSv3	BC, OC, NO_x, NH_3, SO_2, NMVOC, CO, CH_4, CO2	1850-2014	0.5x0.5 degree

Table 1: Characteristics used to develop the 2018 CAMS global emissions

As indicated in D81.2.2, trends for the 2011-2014 period were used to determine the emissions for 2018.

The development of the emissions for 2019 has been changed from what was written in D81.2.2, and we used a more detailed methodology, i.e.:

- Use EDGAR4.3 emissions as a basis for 2010 emissions. EDGAR4.3 provides 0.1x0.1degree gridded emissions for different sectors on a monthly basis

- Calculate the trends in the emissions for 2011-2014 from the last version of CEDS 0.5x0.5 degree emission dataset for each grid point.
- Disaggregate the trends to the same 0.1x0.1 degree grid as EDGAR4.3
- Merge and align sectors between the CEDS and EDGAR4.3 datasets, as indicated in the discussion of Table 2.
- Apply the 2011-2014 trends to the EDGAR4.3 2010 emissions to project emissions for the year 2018. Note that because we do not have data up to the year 2014 for the individual VOCs, a normalized trend calculated for NMVOCs is used to project the emissions for each of the twenty-five individual VOCS.
- Individual VOCs emissions are determined using the VOCs speciation provided by Huang et al. (ACP, 2017) and the JRC group.

Different sectors are available for the 2019 emissions. Since the EDGAR v4.3 sectors and the CEDS sectors do not match, we defined the sectors for the CAMS-GLOB-ANT_v2.1 emissions to allow for the harmonization between the two datasets following Table 2. CAMS-GLOB-ANT_v2.1 emissions are available for 11 sectors, as defined in the first column of the table. Note that these sectors have been created to roughly match the CAMS-REG-AG sectors in order to facilitate the merging of the two datasets in the future. In EDGAR, the sectors are not called the same way for reactive species, methane and VOCs: Table 2 shows the different names used in each inventory for each class of species. The names in parenthesis in the first column corresponds to the sector names in the NetCDF files, as described in deliverable D81.5.2.1-M6. For each sector, the 3rd column for the left shows which CEDS sector was used to determine the recent trends.

CAMS-GLOB-ANT sector	Species in CAMS-GLOB-ANT	CAMS-REG sector	IPCC Sector Name	IPCC Sector
Power Generation (ENE)	All species except vocs	A_PublicPower	Energy Industry	1A1a
Power Generation (ENE)	All vocs		Power generation	1A1
Residential (RCO)	All species	C_OtherStationary Comb	Residential and other sectors	1A4
Road Transportation (TRO)	All species	F_RoadTransport	Road transportation	1A3b
Non-Road Transportation (TNR)	All species	I_OffRoad	Non-road ground transportation	1A3c__1A3e
Fugitive emissions from solid fuels (FEF)	All species except NH3, CO2_excl_sc, monoterpenes (11) and acids (24)	D_Fugitive	Fuel exploitation	1B1a_1B2a1_1B2a2_1B2a3_1B2a4_1B2c and 7A
Fugitive emissions from solid fuels (FEF)	All vocs except monoterpenes (11)		Manufacturing of solid fuels and fugitive emissions	
Industrial processes (IND)	All species except vocs	B_Industry	Oil refineries and transformation	1A1b_1A1c_1A5b1_1B1b_1B2a5_1B2a6_1B2b5_2C1b

Industrial processes (IND)	All species except CO2	B_Industry	Combustion in manufacturing industry	1A2
Industrial processes (IND)	All vocs		Industrial process and product use	2_3
Industrial processes (IND)	All species except NH3, OC and vocs	B_Industry	Iron and steel production	2C1a_2C1c_2C1d _2C1e_2C1f_2C2
Industrial processes (IND)	CO2, CO2_excl_sc, SO2, BC	B_Industry	Aluminum, magnesium and steel production	2C3_2C4_2C5
Industrial processes (IND)	All vocs except vocs monoterpenes (11), esters (18), ethers (19), chlorinated hydrocarbons (20), methanal (21), other alkanals (22), acids (24)		Petroleum refining and distribution	1A1b_1B2a5
Industrial processes (IND)	All species except CO2, OC, and vocs	B_Industry	Chemical processes	2B
Industrial processes (IND)	CO2, CO2_excl_sc	B_Industry	Non-energy use of fuel	2G
Industrial processes (IND)	NMVOC, CO2_excl_sc, NH3, SO2, BC, CO		Non-metallic mineral processes	2A
Industrial processes (IND)	NMVOC, NOx, SO2, BC, OC	B_Industry	Other production	2D
Solvents (SLV)	C02, NH3, NMVOC	E_Solvents	Solvents production and application	3
Agriculture (AGR)	NOx, NH3, CH4, CO, BC, OC and all vocs except alkanols (1), esters (18), chlorinated hydrocarbons (20), acids (24)	L_AgriOther	Agricultural waste burning	4F
Agriculture (AGR)	NOx, NH3, CH4, CO2	L_AgriOther	Agricultural soils	4C_4D1_4D2_4D4
Agriculture (AGR)	CH4	L_AgriOther	Enternic fermentation	4A
Manure Management (MMA)	NOx, NH3, CH4	K_AgriLivestock	Manure management	4B
Ships (SHP)	All species	G_Shipping	Ships	1A3d_1C2
Solid Waste and Waste Water (SWD)	All species except CO2_excl_sc and vocs	J_Waste	Waste incineration	6C
Solid Waste and Waste Water (SWD)	NH3, NMVOC, CH4	J_Waste	Solid waste disposal	6A_6D
Solid Waste and Waste Water (SWD)	NH3, NMVOC, CH4	J_Waste	Waste water	6B

Solid Waste and Waste Water (SWD)	All vocs except esters (18) and ethers (19)		Solid waste and waste water	6

Table 2: Details on the different sectors provided by the CAMS-AG emissions

3. Results: the CAMS-GLOB-ANT_v2.1 2019 inventory

The 2019 global anthropogenic emissions have been developed, as described in the previous section. The emissions are provided on a monthly basis for a large number of species, BC, OC, SO2, CO, NOx, NH3, PM10, PM2.5, as well as NMVOCs and 25 individual VOCs.

As an example of the emissions, Figures 2-4 shows the 2019 emissions for various species and sectors, as well as the differences compared to the 2018 emissions. Between 2018 and 2019, CO emissions decreased over much of central Europe, while increased over parts of southern Europe (notably Spain and Italy), India, China and parts of West Africa (Figure 3). For NOx and SO2, less changes are seen in the last year, with the exception of eastern China which a slight decreasing trend in NOx and slight increasing trend in SO2.

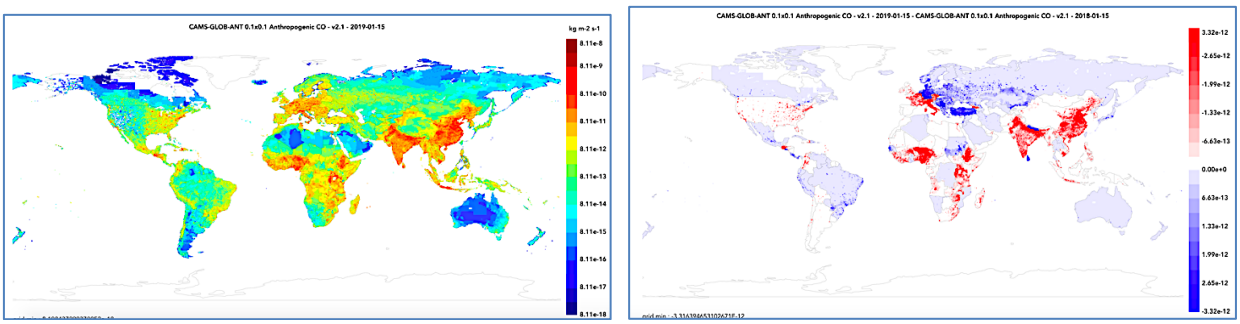


Figure 2: CO emissions in January 2019 from the residential sector (left) and the difference compared to the January 2018 values (right).

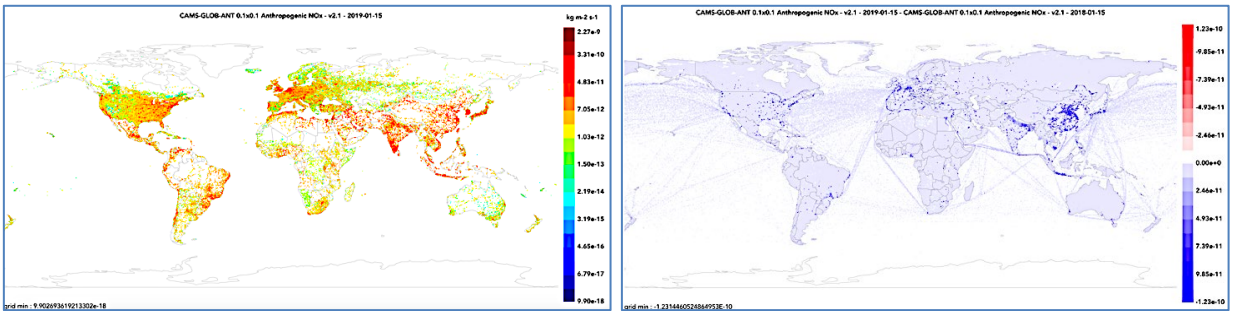


Figure 3: NOx emissions in January 2019 from the road transportation sector (left) and the difference compared to the January 2018 values (right).

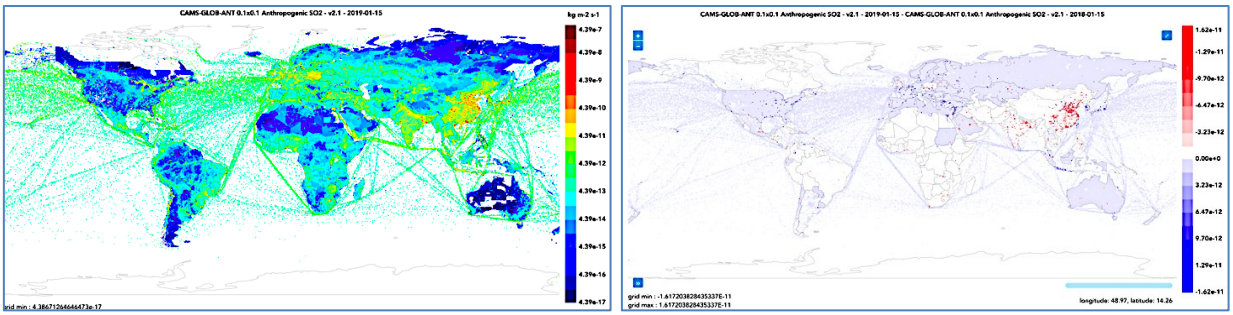


Figure 4: SO2 total emissions in January 2019 from all sectors (left) and the difference compared to the January 2018 values (right).

The list of VOCs available in the CAMS-GLOB-ANT_v2.1 inventory is given in Table 3. The table shows all the sectors available for each compound.

Name	Real name	ENE	RCO	TRO	TNR	FEF	SLV	AGR	SHP	SWD
voc1	Alcohols	X	X	X	X	X	X		X	X
voc2	Ethane	X	X	X	X	X	X	X	X	X
voc3	Propane	X	X	X	X	X	X	X	X	X
voc4	Butanes	X	X	X		X	X	X		X
voc5	Pentanes	X	X	X	X	X	X	X	X	X
voc6	Hexanes	X	X	X	X	X	X	X	X	X
voc7	Ethene	X	X	X	X	X	X	X	X	X
voc8	Propene	X	X	X	X	X	X	X	X	X
voc9	Ethyne	X	X	X	X	X	X	X	X	X
voc10	Isoprenes	X	X	X	X	X	X	X	X	X
voc11	Monoterpenes	X	X	X	X		X	X	X	X
voc12	Other alkad.	X	X	X	X	X	X	X	X	X
voc13	Benzene	X	X	X	X	X	X	X	X	X
voc14	Methylbenzene	X	X	X	X	X	X	X	X	X
voc15	Dimethylbenzenes	X	X	X	X	X	X	X	X	X
voc16	Trimethylbenzenes	X	X	X	X	X	X	X	X	
voc17	Other aromatics	X	X	X	X	X	X	X	X	X
voc18	Esters	X	X	X	X	X	X		X	X
voc19	Ethers	X	X	X	X	X	X	X	X	X
voc20	Chlorinated	X	X	X	X	X	X		X	
voc21	Methanal	X	X	X	X	X	X	X	X	X
voc22	Other alkanals	X	X	X	X	X	X	X	X	X
voc23	Alkanones	X	X	X	X	X	X	X	X	
voc24	Acids	X	X	X	X		X		X	X
voc25	Others	X	X	X	X	X	X	X	X	X

Table 3: List of VOCs considered in the inventory and corresponding sectors

Totals emitted for each compound and for each sector in 2018 are given in Table 4, together with the difference from 2018 to 2019. For the VOCs, totals for 2019 are given.

	ENE	RCO	TRO	TNR	FEF	IND	SLV	AGR	MMA	SHP	SWD	SUM
CO - 2018	6.6	187	165	0.5	3.04	113.9	0.00	77.9	0.00	4.6	0.04	557
Δ 2019	***0.02***	***1.46***	***-3.03***	***-0.02***	***0.02***	***1.67***	***0.00***	***0.00***	***0.00***	***-0.17***	***0.00***	***-0.06***
NO_x - 2018	20.0	3.4	18.4	1.2	0.1	14.2	0.00	4.1	0.3	9.4	0.05	71.2
Δ 2019	***-0.25***	***-0.02***	***-0.27***	***-0.03***	***0.00***	***0.02***	***0.00***	***0.04***	***0.00***	***-0.37***	***0.00***	***-0.68***
SO_2 - 2018	41.9	7.06	0.71	0.06	0.34	36.8	0.00	0.4	0.00	9.1	0.04	97.2
Δ 2019	***-0.84***	***0.09***	***0.01***	***-0.01***	***0.00***	***0.85***	***0.00***	***0.00***	***0.00***	***-0.38***	***0.00***	***-0.28***
OC - 2018	0.12	6.7	0.36	0.03	0.02	1.6	0.00	2.4	0.00	0.18	0.01	11.5
Δ 2019	***0.01***	***0.05***	***0.00***	***0.00***	***0.00***	***0.00***	***0.00***	***0.00***	***0.00***	***-0.01***	***0.00***	***0.04***
BC - 2018	0.14	1.76	0.47	0.03	0.04	1.65	0.00	0.16	0.00	0.35	0.00	4.61
Δ 2019	***0.00***	***0.01***	***0.00***	***0.00***	***0.00***	***0.02***	***0.00***	***0.00***	***0.00***	***-0.01***	***0.00***	***0.02***
CH_4 - 2018	0.35	13.9	0.77	0.01	92.7	8.33	0.00	152.0	12.2	0.06	70.9	351.4
Δ 2019	***0.00***	***0.06***	***0.00***	***0.00***	***1.05***	***0.24***	***0.00***	***1.56***	***0.07***	***0.00***	***0.44***	***3.43***
NH_3 - 2018	0.1	3.95	0.47	0.00	0.00	2.0	0.00	43.1	11.8	2e-3	0.13	61.6
Δ 2019	***0.00***	***0.04***	***-0.01***	***3.4***	***0.00***	***0.03***	***0.00***	***0.41***	***0.07***	***-3.60***	***0.00***	***0.54***
NMVOC - 2018	0.71	25.8	23.4	0.13	26.5	56.9	33.2	4.62	0.00	1.14	2.75	175.
Δ 2019	***0.00***	***0.15***	***-0.82***	***-0.01***	***0.06***	***1.53***	***0.35***	***0.00***	***0.00***	***-0.04***	***-0.01***	***1.21***
VOCs totals emitted for 2019												
alcohols	0.01	0.72	0.29	0	4.31	12.27	0.0	0.0	0.0	0.0	0.51	18.1
ethane	0.01	1.33	0.35	0	2.17	0.39	0.0	0.05	0.0	0.01	0.41	4.71
propane	0.03	0.47	0.25	0.01	3.25	2.01	0.0	0.17	0.0	0.01	0.6	6.26
butanes	0.04	0.06	1.53	0.00	3.22	4.78	0.0	0.04	0.0	0.32	0.01	10.0
pentanes	0.02	0.05	1.58	0.01	2.03	0.65	0.0	0.19	0.0	0.15	0.0	4.68
hexanes	0.05	0.71	7.05	0.05	7.92	4.65	0.0	0.7	0.0	0.11	0.04	21.3
ethene	0.02	2.98	1.51	0.01	0.89	0.76	0.0	0.21	0.0	0.02	0.42	6.81
propene	0.14	0.95	0.87	0.00	0.69	0.49	0.0	0.24	0.0	0.01	0.15	3.42
ethyne	0.01	1.54	0.23	0.0	0.43	0.47	0.0	0.06	0.0	0.01	0.18	2.93
isoprene	0.0	0.0	0.02	0.0	0.3	0.1	0.0	0.08	0.0	0.0	0.0	0.14
monoterpenes	0.0	0.0	0.0	0.0	0.0	0.32	0.0	0.0	0.0	0.01	0.0	0.33
Other alkadienes	0.02	1.52	2.13	0.01	1.14	0.62	0.0	0.37	0.0	0.01	0.26	6.08
benzene	0.05	1.0	0.95	0.00	2.34	0.92	0.O	0.5	0.0	0.06	0.19	6.02
toluene	0.03	0.5	1.4	0.0	2.76	3.40	0.0	0.23	0.0	0.03	0.07	8.43
xylenes	0.04	0.4	1.1	0.00	1.75	4.72	0.0	0.11	0.0	0.01	0.0	8.10
trimethylbenzene	0.0	0.0	0.42	0.0	0.0	0.39	0.0	0.05	0.0	0.0	0.0	0.86
aromatics	0.01	2.1	1.97	0.01	1.31	1.91	0.0	0.09	0.0	0.02	0.1	7.46
esters	0.01	0.1	0.0	0.0	0.50	1.02	0.0	0.0	0.0	0.0	0.0	1.62
ethers	0.0	0.1	0.0	0.0	1.49	0.99	0.0	0.01	0.0	0.0	0.0	2.59
Chlorinated	0.09	0.03	0.0	0.01	3.66	3.07	0.0	0.0	0.0	0.0	0.0	6.86
formaldehyde	0.16	1.0	0.17	0.00	2.82	0.43	0.0	0.14	0.0	0.01	0.01	4.76
aldehydes	0.01	6.0	0.17	0.0	2.92	0.50	0.0	0.45	0.0	0.20	0.08	10.37
ketones	0.03	1.6	0.05	0.0	3.87	3.65	0.0	0.05	0.0	0.06	0.03	9.35
alkanones	0.01	0.2	0.00	0.0	9.7	0.85	0.0	0.0	0.0	0.01	0.0	10.85
others	0.01	2.2	0.60	0.0	1.48	9.06	0.0	0.93	0.0	0.01	0.21	14.49

Table 4: top: totals emitted in 2018 for the different sectors, together with the difference between 2018 and 2019; below: totals emitted for VOCs for different sectors, for 2019.

Emissions for 2019 will be made available to all CAMS users in the following weeks, through the ECCAD database.

4. Methodology for developing CAMS-GLOB-AIR, the aircraft emissions

The CAMS-GLOB-AIR_v1.1 dataset contains aircraft emissions on a 0.5° X 0.5° horizontal grid for 25 levels, covering the period 2000-2019. The emissions are based on the CEDS aircraft emission data as described in Hoesly et al. (GMD, 2017). For the years up to 2014, the emissions are the same as CEDS. After 2014, we extrapolate in time using the trends calculated for the period 2012-2014. These dates were chosen because the trends are more stable after 2011, which may be a reflection of the global economic crisis. For the speciation of VOCs, we use weights defined by EDGAR for landing and taking (for the first two levels of the atmosphere corresponding to 0.305 km and 0.915 km), and for exhaust (corresponding to the rest of the levels up to 14.945 km). The emission for each individual VOC is calculated by multiplying the weight as given in Table 5 by the emissions for total VOCs.

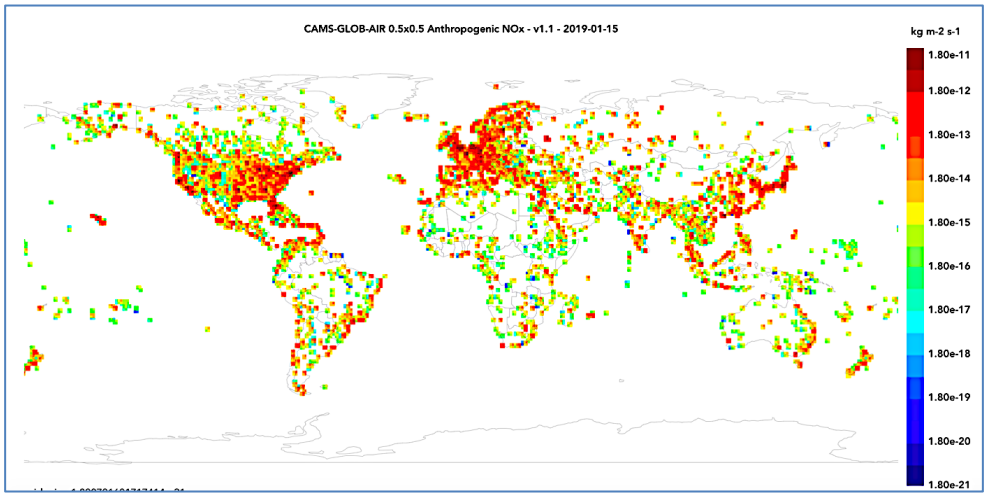
VOC species	**Wt for exhaust**	**Wt for landing/takeoff**
1-alcohols (alkanols)	0.0180509	0.02032566
2-ethane	0.00521026	0.01033127
3-propane	0.00078004	0.00224593
4-butanes	0.0	0.0
5-pentanes	0 .0019801	0.00213363
6-hexanes	0.1811991	0.1066816
7-ethene	0.1546177	0.1740595
8-propene	0.04534227	0.05154408
9-ethyne	0.03939197	0.04143739
10-isoprenes	0.0	0.0
11-monoterpenes	0.0	0.0
12-olef (other alkenes/alkynes)	0.1335667	0.08175182
13-benzene	0.01681084	0.02032566
14-toluene (methylbenzene)	0.00642032	0.00550253
15-xylenes (dimethylbenzenes)	0.00448022	0.00494104
16-trimethylbenzenes	0.00510026	0.0
17-arom	0.04079204	0.02930938
18-esters	0.0	0.0
19-ethers	0.0	0.0

20-clhyd	0.0	0.0
21-formaldehyde (methanal)	0.1231062	0.1587872
22-aldehydes (other alkanals)	0.219461	0.125772
23-ketones (alkanones)	0.00369018	0.03290286
24-acids	0.0	0.0
25-other	0.0	0.1321729

Table 5: Weights describing the speciation for aircraft emission as taken from EDGAR. Takeoff landing corresponds to the lowest two layers (0.305 km and 0.915 km). See http://www.globalchange.umd.edu/data/ceds/README-CEDS-VOC-speciation_2017-05-18.txt

5. The CAMS-GLOB-AIR aircraft emissions

Maps of CO and NOx aircraft emissions from the lowest level (centered at 0.305 km) and near cruising altitude (centered at 10.065 km) are presented in Figures 5 and 6. Near the surface, and the lowest level, the emissions represent the landing and taking of aircraft and are thus concentrated near the airports. At cruising altitudes, emissions are concentrated along the aircraft flight paths.



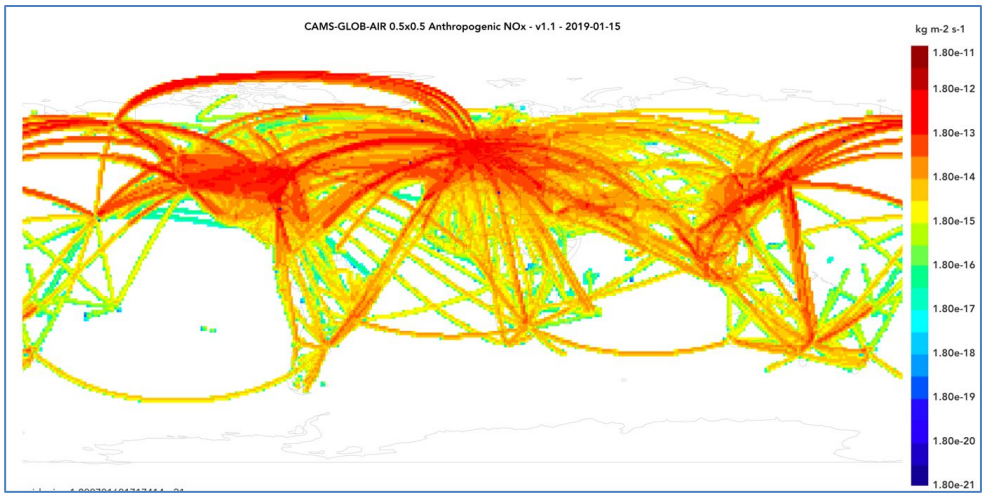


Figure 5: Aircraft emissions of NOx for January 2019 at the level centered at 0.305 km (top) and the level centered at 10.065 km (bottom).

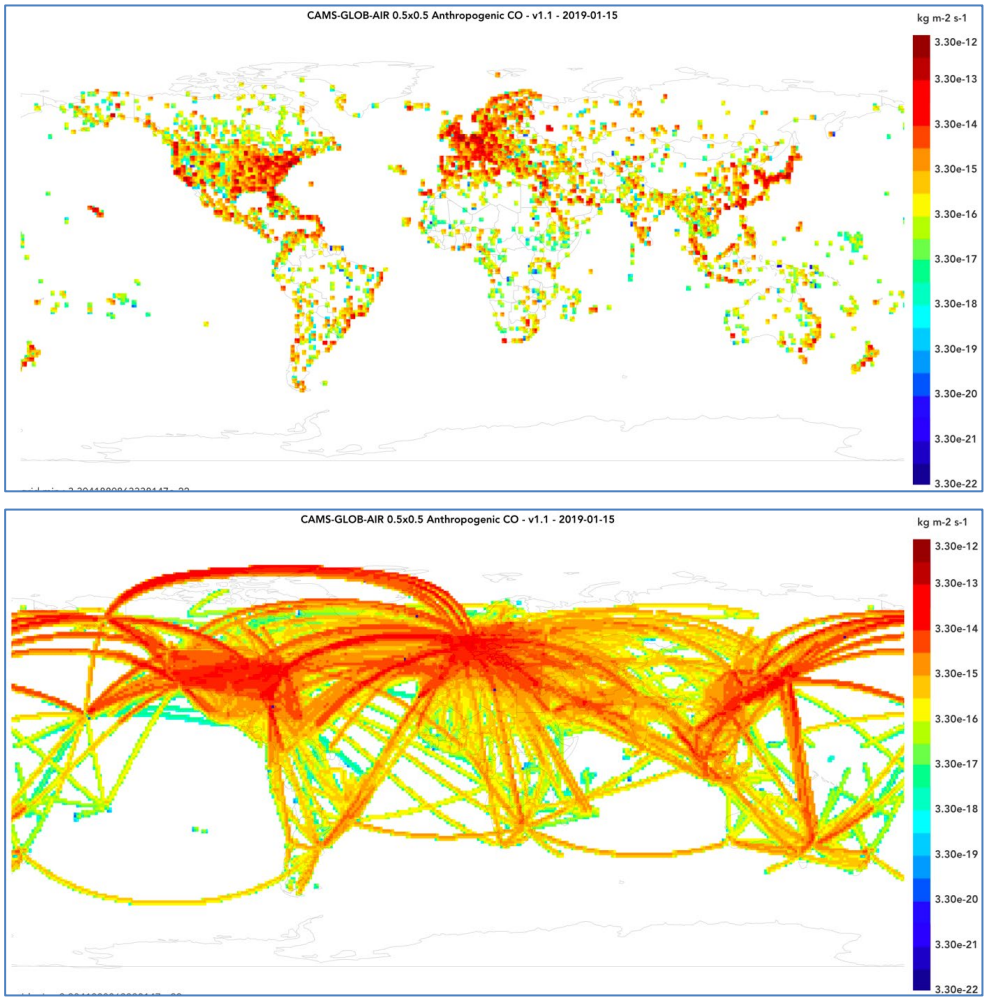


Figure 6: Aircraft emissions of CO for January 2019

Copernicus Atmosphere Monitoring Service

ECMWF - Shinfield Park, Reading RG2 9AX, UK

Contact: info@copernicus-atmosphere.eu

atmosphere.copernicus.eu copernicus.eu ecmwf.int